FGFR2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR2 gene fusions from the msk_impact_50k_2026 study, covering 136 fusion events. 108/116 fusions with known frame status (93.1%) were in-frame. 115/134 fusions (85.8%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.00226743, statistically significant at alpha=0.05). No literature benchmark is configured for FGFR2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 136 analyzed fusion events, identifying 78 distinct fusion partner genes. The most frequent partner was BICC1, observed in 28 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 99/108 (91.7%) of in-frame fusions retained the domain, p=0.00226743 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Disruption of the C-terminal autoinhibitory tail (exon 18) was tested with Fisher's exact test comparing in-frame fusions against all others; 88/95 (92.6%) of in-frame fusions disrupted the domain, p=0.154955 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.386139.

Junction concentration

93% of 108 fusions with this gene as the 5' partner join at exon 17, against 3% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 119 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 767 aa (exon 17; label-permutation p=0.000999001). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 10 aa from the nearest configured domain boundary at 757 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=104) and "not_retained" (n=15). For Frame_status, retained: 88/91 (96.7%, 95% CI 90.8%-98.9%); not_retained: 7/12 (58.3%, 95% CI 32.0%-80.7%). Welch's t-test comparing Tumor_variant_count between groups gave means of 107.933 vs 65.5333, p=0.262021 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (78 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for FGFR2 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (9 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 3 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: FGFR2 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run (7 row(s) in retention_counter_intuitive_events, 1 row(s) in retention_counter_intuitive_recurrent_partners); see the accompanying table.

Mutation cooccurrence

Mutation cooccurrence produced results for this run (1 row(s) in cooccurrence_results); see the accompanying table.

Window detection

Window detection produced results for this run (32 row(s) in window_scan, 19 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BICC1	28	0.20588235294117646	0.5000004342923104	0.041325676675614334	0.9965603502188868	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.41609165050070474	1
TACC2	8	0.058823529411764705	0.5000004342923104	0.041325676675614334	0.991869918699187	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3712704387139262	2
KIAA1217	4	0.029411764705882353	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.36366642149728345	3
NOL4	4	0.029411764705882353	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.36366642149728345	4
CTNNA3	3	0.022058823529411766	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3614605391443423	5
SHTN1	3	0.022058823529411766	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3614605391443423	6
BAIAP2L1	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3592546567914011	7
NRBF2	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3592546567914011	8
OPTN	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3592546567914011	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SORBS1	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3592546567914011	10
TTC28	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3592546567914011	11
WAC	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3592546567914011	12
AHCYL1	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	0.9991869918699187	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3591327055718889	13
INA	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	0.9983739837398374	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3590107543523767	14
ABLIM1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	15
ADAM9	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	16
AMOT	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	17
ATAD1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	18
ATP6V1D	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BICD1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	20
BNIP2	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	21
BTBD16	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	22
CAT	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	23
CCDC186	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	24
CCDC6	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	25
CCDC73	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	26
CEACAM7	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	27
DDX21	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	28
GAB2	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
HSPA12A	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	30
KIF14	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	31
L3MBTL3	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	32
LAMC1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	33
MARVELD3	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	34
MBIP	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	35
MYH15	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	36
MYH9	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	37
MYLK	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	38
MYO18B	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NRAP	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	40
PAH	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	41
PALLD	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	42
PDZRN4	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	43
PKP4	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	44
POC1B	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	45
PPP1R21	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	46
PRDX3	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	47
ROCK1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	48
SEPTIN10	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SH3GLB1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	50
SHC2	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	51
SKI	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	52
SRRT	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	53
STAT4	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	54
TFEC	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	55
UBP1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	56
VCL	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	57
ZFAT	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	58
ZMYM4	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	1.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35704877443845995	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CEP112	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.9983739837398374	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35680487199943556	60
NID2	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.9983739837398374	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35680487199943556	61
RNF145	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.9983739837398374	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35680487199943556	62
RPAP3	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.9983739837398374	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35680487199943556	63
VPS26A	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.9983739837398374	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35680487199943556	64
FOXP1	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	0.9723577235772358	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3551083153279865	65
NRG3	2	0.014705882352941176	0.5000004342923104	0.041325676675614334	0.9707317073170731	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3548644128889621	66
KCNH7	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.9495934959349593	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.34948779882870384	67
ANKRD11	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.943089430894309	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3485121890726063	68
ATE1	3	0.022058823529411766	0.5000004342923104	0.041325676675614334	0.8384823848238483	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3372328968679196	69

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ADAM12	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.8520325203252033	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3348536524872404	70
TBC1D4	3	0.022058823529411766	0.5000004342923104	0.041325676675614334	None	0.7157764817731829	confidence_certainty, domain_disruption, domain_retention, recurrence	0.24877710487569682	71
DIABLO	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	None	0.7157764817731829	confidence_certainty, domain_disruption, domain_retention, recurrence	0.24358679345701173	72
KLHL29	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	None	0.7157764817731829	confidence_certainty, domain_disruption, domain_retention, recurrence	0.24358679345701173	73
PRKAG2	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	None	0.7157764817731829	confidence_certainty, domain_disruption, domain_retention, recurrence	0.24358679345701173	74
RABGAP1L	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	None	0.7157764817731829	confidence_certainty, domain_disruption, domain_retention, recurrence	0.24358679345701173	75
ST8SIA6	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	None	0.7157764817731829	confidence_certainty, domain_disruption, domain_retention, recurrence	0.24358679345701173	76
YTHDF3	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	None	0.7157764817731829	confidence_certainty, domain_disruption, domain_retention, recurrence	0.24358679345701173	77
GRID1	1	0.007352941176470588	0.5000004342923104	0.041325676675614334	0.0	0.7157764817731829	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.20704877443845995	78

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
152	0	104	1	14	0.0	0.12605042016806725	0.8994557023368495
469	0	104	2	13	0.0	0.014955134596211367	1.8252096739647368
676	0	104	3	12	0.0	0.0016616816218012625	2.7794521834040617
732	0	104	5	10	0.0	1.6442426842411296e-05	4.784034081778742

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
767	100	4	5	10	50.0	2.37761309478641e-08	7.623858816021042
768	100	4	15	0	0.0	1.0	-0.0
798	101	3	15	0	0.0	1.0	-0.0
801	102	2	15	0	0.0	1.0	-0.0
802	103	1	15	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

88	20
7	4

frame_domain_contingency_tables_by_tumor_type

stratum	table
ADNOS	[6, 0], [0, 0]
BLCA	[0, 1], [0, 0]
BRCA	[0, 1], [0, 0]
BRCANOS	[0, 0], [0, 0]
CEAS	[1, 0], [0, 0]
CHOL	[8, 0], [0, 0]
COAD	[2, 3], [0, 0]
CUP	[3, 1], [1, 0]
CUPNOS	[1, 1], [0, 0]
EGC	[1, 0], [0, 0]
ESCA	[2, 0], [0, 0]
GB	[1, 0], [0, 0]
GNG	[1, 0], [0, 0]
HCCIHCH	[1, 0], [0, 0]
HGSOC	[1, 1], [0, 0]
IDC	[4, 2], [1, 3]
IHCH	[43, 7], [4, 0]
LGSOC	[1, 0], [0, 0]
LUAD	[3, 0], [0, 0]
LUSC	[0, 0], [0, 1]

stratum	table
PAAD	[3, 0], [0, 0]
PAMPCA	[1, 0], [0, 0]
PLLS	[0, 1], [0, 0]
PRAD	[0, 1], [0, 0]
SARCL	[1, 0], [0, 0]
SPDAC	[1, 0], [0, 0]
STAD	[1, 0], [0, 0]
SYNS	[0, 0], [1, 0]
UEC	[1, 0], [0, 0]
USC	[1, 1], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
c_terminal_tail	C-terminal autoinhibitory tail (exon 18)	119	11	4	104	0.02623456790123457

domain_retention tables

frame_domain_contingency_table

88	16
7	8

frame_domain_contingency_tables_by_tumor_type

stratum	table
ADNOS	[6, 0], [0, 0]
BLCA	[0, 1], [0, 0]
BRCA	[0, 1], [0, 0]
CEAS	[1, 0], [0, 0]
CHOL	[8, 0], [0, 0]
COAD	[2, 3], [0, 0]
CUP	[3, 1], [1, 0]
CUPNOS	[1, 1], [0, 0]
EGC	[1, 0], [0, 0]
ESCA	[2, 0], [0, 0]
GB	[1, 0], [0, 0]

stratum	table
GNG	[1, 0], [0, 0]
HCCIHCH	[1, 0], [0, 0]
HGSOC	[1, 0], [0, 1]
IDC	[4, 1], [1, 4]
IHCH	[43, 7], [4, 0]
LGSOC	[1, 0], [0, 0]
LUAD	[3, 0], [0, 0]
LUSC	[0, 0], [0, 1]
PAAD	[3, 0], [0, 0]
PAMPCA	[1, 0], [0, 0]
PLLS	[0, 0], [0, 1]
PRAD	[0, 0], [0, 1]
SARCL	[1, 0], [0, 0]
SPDAC	[1, 0], [0, 0]
STAD	[1, 0], [0, 0]
SYNS	[0, 0], [1, 0]
UEC	[1, 0], [0, 0]
USC	[1, 1], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	119	104	3	12	0.11407942238267148

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	106	121	0.8760330578512396

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ABLIM1	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ADAM12	1	1	1	1
ADAM9	1	1	1	1
AHCYL1	2	2	2	2
AMOT	1	1	1	1
ANKRD11	1	1	1	1
ATAD1	1	1	1	1
ATE1	3	3	3	3
ATP6V1D	1	1	1	1
BAIAP2L1	2	2	2	2
BICC1	28	28	28	26
BICD1	1	1	1	1
BNIP2	1	1	1	1
BTBD16	1	1	1	1
CAT	1	1	1	1
CCDC186	1	1	1	1
CCDC6	1	1	1	1
CCDC73	1	1	1	1
CEACAM7	1	1	1	1
CEP112	1	1	1	1
CTNNA3	3	3	3	3
DDX21	1	1	1	1
DIABLO	1	1	1	1
FOXP1	2	2	2	2
GAB2	1	1	1	1
GRID1	1	1	1	1
HSPA12A	1	1	1	1
INA	2	2	2	1
KCNH7	1	1	1	1
KIAA1217	4	4	4	4
KIF14	1	1	1	1
KLHL29	1	1	1	1
L3MBTL3	1	1	1	1
LAMC1	1	1	1	1
MARVELD3	1	1	1	1
MBIP	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
MYH15	1	1	1	1
MYH9	1	1	1	1
MYLK	1	1	1	1
MYO18B	1	1	1	1
NID2	1	1	1	1
NOL4	4	4	4	3
NRAP	1	1	1	1
NRBF2	2	2	2	1
NRG3	2	2	2	2
OPTN	2	2	2	1
PAH	1	1	1	1
PALLD	1	1	1	1
PDZRN4	1	1	1	1
PKP4	1	1	1	1
POC1B	1	1	1	1
PPP1R21	1	1	1	1
PRDX3	1	1	1	1
PRKAG2	1	1	1	1
RABGAP1L	1	1	1	1
RNF145	1	1	1	1
ROCK1	1	1	1	1
RPAP3	1	1	1	1
SEPTIN10	1	1	1	1
SH3GLB1	1	1	1	1
SHC2	1	1	1	1
SHTN1	3	3	3	3
SKI	1	1	1	1
SORBS1	2	2	2	2
SRRT	1	1	1	1
ST8SIA6	1	1	1	1
STAT4	1	1	1	1
TACC2	8	8	8	7
TBC1D4	3	3	3	1
TFEC	1	1	1	1
TTC28	2	2	2	2

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
UBP1	1	1	1	1
VCL	1	1	1	1
VPS26A	1	1	1	1
WAC	2	2	2	2
YTHDF3	1	1	1	1
ZFAT	1	1	1	1
ZMYM4	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
10	123240000	123241000	32	EVT-P-0002810-T01-IM3-169, EVT-P-0003660-T01-IM5-110, EVT-P-0004520-T01-IM5-55, EVT-P-0006235-T01-IM5-188, EVT-P-0006885-T01-IM5-41, ... (32 total)
10	123242000	123243000	32	EVT-P-0003177-T01-IM5-5, EVT-P-0004089-T01-IM5-158, EVT-P-0005538-T01-IM5-7, EVT-P-0008693-T01-IM5-178, EVT-P-0012423-T01-IM5-176, ... (32 total)
10	123241000	123242000	27	EVT-P-0002675-T01-IM3-4, EVT-P-0003724-T01-IM5-20, EVT-P-0005291-T01-IM5-153, EVT-P-0006328-T01-IM5-179, EVT-P-0008658-T01-IM5-108, ... (27 total)
10	123239000	123240000	13	EVT-P-0000644-T01-IM3-30, EVT-P-0009701-T01-IM5-105, EVT-P-0015500-T01-IM6-122, EVT-P-0019439-T02-IM6-112, EVT-P-0020135-T01-IM6-58, ... (13 total)
10	123243000	123244000	11	EVT-P-0000386-T02-IM5-170, EVT-P-0011516-T01-IM5-132, EVT-P-0019723-T01-IM6-23, EVT-P-0020410-T01-IM6-24, EVT-P-0021826-T01-IM6-1, ... (11 total)
10	123244000	123245000	1	EVT-P-0049071-T01-IM6-180
10	123246000	123247000	1	EVT-P-0026275-T01-IM6-27
10	123263000	123264000	1	EVT-P-0063512-T01-IM7-36
10	123323000	123324000	1	EVT-P-0055215-T01-IM6-103

exact_position_recurrence

chromosome	position	n_events	event_ids
10	123240418	2	EVT-P-0024094-T01-IM6-172, EVT-P-0038235-T01-IM6-119

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
767	100	99	3528	False
768	10	10	3382	False
732	2	2	1083	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	17	100	108	0.9259259259259259	0.03158404596472475	138.61703095479612
three_prime	18	10	11	0.9090909090909091	0.044737105825761625	12.469960483641422
five_prime	18	4	108	0.037037037037037035	0.014038281667557455	1.1782078881184466
five_prime	4	1	108	0.009259259259259259	0.10956707642971673	1.5657258716466828e-06
five_prime	10	1	108	0.009259259259259259	0.025003340459647246	0.029148405334171328
five_prime	15	1	108	0.009259259259259259	0.015800374131480492	0.08568564835802239
five_prime	16	1	108	0.009259259259259259	0.01443913682522715	0.10120897125707576
three_prime	17	1	11	0.09090909090909091	0.014171900053447354	0.8377309736433594

mechanistic_interpretation tables

retention_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0037617-T01-IM6-3	P-0037617-T01-IM6	BICC1	768	lost
EVT-P-0025250-T01-IM6-10	P-0025250-T01-IM6	BICC1	768	lost
EVT-P-0003724-T01-IM5-20	P-0003724-T01-IM5	CEP112	768	lost
EVT-P-0000644-T01-IM3-30	P-0000644-T01-IM3	AHCYL1	768	lost
EVT-P-0013756-T01-IM5-166	P-0013756-T01-IM5	INA	768	lost
EVT-P-0039370-T01-IM6-189	P-0039370-T01-IM6	RPAP3	768	lost
EVT-P-0021788-T01-IM6-193	P-0021788-T01-IM6	VPS26A	768	lost

retention_counter_intuitive_recurrent_partners

partner_gene	count
BICC1	2

mutation_cooccurrence tables

cooccurrence_results

comparator_gene	alteration_type	protein_change	label	contingency_table	odds_ratio	p_value	direction	cohort_sample_count	supplied_cohort_sample_count	excluded_cohort_sample_count	excluded_fusion_positive_sample_count	excluded_comparator_sample_count	counting_unit	eligibility_source	eligible_sample_ids_sha256	fusion_positive_sample_count	comparator_altered_sample_count	both_altered_sample_count
FGFR2	point_mutation	S252W	FGFR2 point_mutation (S252W)	[0, 121], [108, 54102]	0.0	1.0	mutually_exclusive	54331	54331	0	0	0	sample	joint_assay_metadata	b3ed60b4597be585f87974d350014c979cdf1535fa97f1a63f227a7531ea7127	121	108	0

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
567	767	200	103	16	100	4	100.0	7.847464893592361e-11	10.105270618482743	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, EVT-P-0003660-T01-IM5-110, ... (103 total)
568	768	200	113	6	100	4	3.8461538461538463	0.1652077618639387	0.7819695523213918	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
598	798	200	114	5	101	3	5.17948717948718	0.11883041530082415	0.9250723850561248	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (114 total)
601	801	200	115	4	102	2	7.846153846153846	0.07695304889927918	1.1137741686343026	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (115 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
667	767	100	103	16	100	4	100.0	7.847464893592361e-11	10.105270618482743	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, EVT-P-0003660-T01-IM5-110, ... (103 total)
668	768	100	113	6	100	4	3.8461538461538463	0.1652077618639387	0.7819695523213918	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
676	776	100	113	6	100	4	3.8461538461538463	0.1652077618639387	0.7819695523213918	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
698	798	100	113	6	101	3	8.416666666666666	0.026075722174595126	1.5837636548247884	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
701	801	100	114	5	102	2	12.75	0.014136999296961692	1.8496427635991088	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (114 total)
702	802	100	115	4	103	1	25.75	0.006131032190783971	2.212466403617808	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (115 total)
717	767	50	102	17	100	4	162.5	2.97855935452814e-12	11.525993741179636	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, EVT-P-0003660-T01-IM5-110, ... (102 total)
718	768	50	112	7	100	4	6.25	0.04208206302381164	1.3759029775545222	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (112 total)
732	782	50	112	7	100	4	6.25	0.04208206302381164	1.3759029775545222	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (112 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
742	767	25	100	19	100	4	None	9.3641107738865e-16	15.028533457405668	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, EVT-P-0003660-T01-IM5-110, ... (100 total)
743	768	25	110	9	100	4	12.5	0.0015254561423508797	2.816600274031936	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (110 total)
748	798	50	111	8	101	3	16.833333333333332	0.0007323491633356469	3.135281810259301	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (111 total)
751	801	50	112	7	102	2	25.5	0.0002965329520946031	3.5279270388336923	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (112 total)
752	802	50	113	6	103	1	51.5	9.144297033411193e-05	4.038849675233482	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
767	792	25	110	9	100	4	12.5	0.0015254561423508797	2.816600274031936	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (110 total)
767	817	50	113	6	103	1	51.5	9.144297033411193e-05	4.038849675233482	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
767	867	100	114	5	104	0	None	1.6442426842411296e-05	4.784034081778742	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (114 total)
767	967	200	114	5	104	0	None	1.6442426842411296e-05	4.784034081778742	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (114 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
768	793	25	10	109	0	104	0.0	2.8224790277751077e-11	10.54936927638453	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0013756-T01-IM5-166, EVT-P-0021788-T01-IM6-193, EVT-P-0021975-T01-IM6-175, ... (10 total)
768	818	50	13	106	3	101	0.01485148514851485	7.09640239619849e-09	8.148961766133816	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0013756-T01-IM5-166, EVT-P-0021788-T01-IM6-193, ... (13 total)
768	868	100	14	105	4	100	0.02	2.3776130947864096e-08	7.623858816021042	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0013756-T01-IM5-166, EVT-P-0021788-T01-IM6-193, ... (14 total)
768	968	200	14	105	4	100	0.02	2.3776130947864096e-08	7.623858816021042	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0013756-T01-IM5-166, EVT-P-0021788-T01-IM6-193, ... (14 total)
770	820	50	4	115	4	100	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-146, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-51
795	820	25	4	115	4	100	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-146, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-51
798	823	25	4	115	4	100	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-146, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-51
798	848	50	4	115	4	100	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-146, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-51
798	898	100	4	115	4	100	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-146, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-51
798	998	200	4	115	4	100	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-146, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-51

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
742	767	25	100	19	100	4	None	9.3641107738865e-16	15.028533457405668	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, EVT-P-0003660-T01-IM5-110, ... (100 total)
717	767	50	102	17	100	4	162.5	2.97855935452814e-12	11.525993741179636	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, EVT-P-0003660-T01-IM5-110, ... (102 total)
768	793	25	10	109	0	104	0.0	2.8224790277751077e-11	10.54936927638453	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0013756-T01-IM5-166, EVT-P-0021788-T01-IM6-193, EVT-P-0021975-T01-IM6-175, ... (10 total)
667	767	100	103	16	100	4	100.0	7.847464893592361e-11	10.105270618482743	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, EVT-P-0003660-T01-IM5-110, ... (103 total)
768	818	50	13	106	3	101	0.01485148514851485	7.09640239619849e-09	8.148961766133816	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0013756-T01-IM5-166, EVT-P-0021788-T01-IM6-193, ... (13 total)
768	868	100	14	105	4	100	0.02	2.3776130947864096e-08	7.623858816021042	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0013756-T01-IM5-166, EVT-P-0021788-T01-IM6-193, ... (14 total)
767	867	100	114	5	104	0	None	1.6442426842411296e-05	4.784034081778742	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (114 total)
752	802	50	113	6	103	1	51.5	9.144297033411193e-05	4.038849675233482	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
751	801	50	112	7	102	2	25.5	0.0002965329520946031	3.5279270388336923	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (112 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
748	798	50	111	8	101	3	16.833333333333332	0.0007323491633356469	3.135281810259301	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (111 total)
743	768	25	110	9	100	4	12.5	0.0015254561423508797	2.816600274031936	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (110 total)
702	802	100	115	4	103	1	25.75	0.0061310321907839 71	2.212466403617808	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (115 total)
701	801	100	114	5	102	2	12.75	0.014136999296961692	1.8496427635991088	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (114 total)
698	798	100	113	6	101	3	8.416666666666666	0.026075722174595126	1.5837636548247884	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)
718	768	50	112	7	100	4	6.25	0.04208206302381164	1.3759029775545222	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (112 total)
601	801	200	115	4	102	2	7.846153846153846	0.07695304889927918	1.1137741686343026	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (115 total)
598	798	200	114	5	101	3	5.17948717948718	0.11883041530082415	0.9250723850561248	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (114 total)
668	768	100	113	6	100	4	3.8461538461538463	0.1652077618639387	0.7819695523213918	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-5, ... (113 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
795	820	25	4	115	4	100	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-146, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-51

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021826-T01-IM6-1	P-0021826-T01-IM6	P-0021826	Breast Cancer	IDC	ANKRD11-FGFR2	ANKRD11	unknown	five_prime	123243799	16	732	True	731	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.909747292418 7726	True		C-terminal autoinhibitory tail (exon 18)	Protein kinase domain	FGFR2 (NM_000141) - ANKRD11 (NM_013275) rearrangement: t(10,16)(q26.13,q24.3)(chr10:g.123243799::chr16:g.89538175) | Transcript Fusion { FGFR2:ANKRD11}	NA
EVT-P-0013241-T01-IM5-2	P-0013241-T01-IM5	P-0013241	Hepatobiliary Cancer	IHCH	ATP6V1D-FGFR2	ATP6V1D	in-frame	five_prime	123242048	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ATP6V1D (NM_015994) rearrangement: t(10;14)(q26.13;q23.2)(chr10:g.123242048::chr14:g.67818116) | Protein fusion: in frame (FGFR2-ATP6V1D)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037617-T01-IM6-3	P-0037617-T01-IM6	P-0037617	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	three_prime	123241302	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		BICC1 (NM_001080512) - FGFR2 (NM_000141) rearrangement: c.238-34060:BICC1_c.2302-1767:FGFR2inv | Protein Fusion: in frame {BICC1: FGFR2}	NA
EVT-P-0002675-T01-IM3-4	P-0002675-T01-IM3	P-0002675	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241096	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		NA | Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0003177-T01-IM5-5	P-0003177-T01-IM5	P-0003177	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123242188	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		NA | Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0008146-T01-IM5-6	P-0008146-T01-IM5	P-0008146	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123240154	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2302-619:FGFR2_c.2182-744:BICC1inv | Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005538-T01-IM5-7	P-0005538-T01-IM5	P-0005538	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123242753	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2301+459:FGFR2_c.2182-276:BICC1inv | Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009512-T01-IM5-8	P-0009512-T01-IM5	P-0009512	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123240449	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		Note: The FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal rearrangement event is an inversion which results in the fusion of FGFR2 exons 1-17 with BICC1 exons 3-21 . The fusion is predicted to be in frame and includes the kinase domain of FGFR2. | Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015942-T01-IM6-9	P-0015942-T01-IM6	P-0015942	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241020	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exons 1-17 with BICC1 exons 9-21) : c.2302-1485:FGFR2_c.1048-60:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0025250-T01-IM6-10	P-0025250-T01-IM6	P-0025250	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	three_prime	123240491	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 16-21): c.2302-956:FGFR2_c.2182-297:BICC1inv | Protein Fusion: in frame {BICC1:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020025-T01-IM6-11	P-0020025-T01-IM6	P-0020025	Cancer of Unknown Primary	ADNOS	BICC1-FGFR2	BICC1	in-frame	five_prime	123241984	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 3-21) : c.2301+1228:FGFR2_c.238-4773:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0020186-T01-IM6-12	P-0020186-T01-IM6	P-0020186	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241148	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exon17 fused with BICC1 exon3): c.2302-1613:FGFR2_c.237+32254:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029067-T01-IM6-13	P-0029067-T01-IM6	P-0029067	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123240197	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 3-21) : c.2302-67:FGFR2_c.238-6757:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0018115-T01-IM6-14	P-0018115-T01-IM6	P-0018115	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241970	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused in-frame with BICC1 exons 3-21): c.2301+1242:FGFR2_c.237+34947:BICC1dup | Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032033-T01-IM6-15	P-0032033-T01-IM6	P-0032033	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241685	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exon 18 fused with BICC1 exon3) : c.2301+1527:FGFR2_c.237+38877:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0025807-T01-IM6-16	P-0025807-T01-IM6	P-0025807	Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123242181	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 3-21): c.2301+895:FGFR2_c.238-35107:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015432-T01-IM6-17	P-0015432-T01-IM6	P-0015432	Pancreatic Cancer	PAAD	CAT-FGFR2	CAT	in-frame	five_prime	123242534	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - CAT (NM_001752) rearrangement: t(10;11)(q26.13;p13)(chr10:g.123242534::chr11:g.34467980) | Protein fusion: in frame (FGFR2-CAT)	NA
EVT-P-0027305-T01-IM6-18	P-0027305-T01-IM6	P-0027305	Esophagogastric Cancer	STAD	CCDC6-FGFR2	CCDC6	in-frame	five_prime	123241542	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - CCDC6 (NM_005436) fusion (FGFR2 exons 1-17 fused to CCDC6 exons 4-9): c.2301+1670:FGFR2_c.582+6426:CCDC6del | Protein Fusion: in frame {FGFR2:CCDC6}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027741-T01-IM6-19	P-0027741-T01-IM6	P-0027741	Breast Cancer	IDC	CEACAM7-FGFR2	CEACAM7	unknown	five_prime	123241990	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) rearrangement: t(10;19)(q26.13;q13.2)(chr10:g.123241990::chr19:g.42175729) | Transcript Fusion { FGFR2:CEACAM7}	NA
EVT-P-0003724-T01-IM5-20	P-0003724-T01-IM5	P-0003724	Soft Tissue Sarcoma	SYNS	CEP112-FGFR2	CEP112	in-frame	three_prime	123241923	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		t(17;10)(q24.1;q26.13)(chr17:g.64023428::chr10:g.123241923) | Protein fusion: in frame (CDC46-FGFR2)	NA
EVT-P-0011516-T01-IM5-21	P-0011516-T01-IM5	P-0011516	Hepatobiliary Cancer	IHCH	CTNNA3-FGFR2	CTNNA3	in-frame	three_prime	123241940	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		CTNNA3 (NM_013266) -FGFR2 (NM_000141) Rearrangement : c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup | Protein fusion: in frame (CTNNA3-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019723-T01-IM6-23	P-0019723-T01-IM6	P-0019723	Esophagogastric Cancer	SPDAC	CTNNA3-FGFR2	CTNNA3	in-frame	five_prime	123243018	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del | Protein Fusion: in frame {FGFR2:CTNNA3}	NA
EVT-P-0020410-T01-IM6-24	P-0020410-T01-IM6	P-0020410	Hepatobiliary Cancer	IHCH	DDX21-FGFR2	DDX21	in-frame	five_prime	123243048	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - DDX21 (NM_004728) fusion (FGFR2 exons 1-17 fused to DDX21 exons 2-15): c.2301+164:FGFR2_c.88-860:DDX21inv | Protein Fusion: in frame {FGFR2:DDX21}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018869-T03-IM6-26	P-0018869-T03-IM6	P-0018869	Breast Cancer	IDC	FGFR2-ABLIM1	ABLIM1	in-frame	five_prime	123242601	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ABLIM1 (NM_002313) fusion: c.2301+611:FGFR2_c.895-12244:ABLIM1del | Protein Fusion: in frame {FGFR2:ABLIM1}	NA
EVT-P-0026275-T01-IM6-27	P-0026275-T01-IM6	P-0026275	Soft Tissue Sarcoma	PLLS	FGFR2-ADAM12	ADAM12	unknown	five_prime	123246899	15	676	False	675	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.707581227436231	True		C-terminal autoinhibitory tail (exon 18)	Protein kinase domain	FGFR2 (NM_000141) - ADAM12 (NM_003474) rearrangement: c.2026:FGFR2_c.1618+5260:ADAM12dup | Protein Fusion: mid-exon {FGFR2:ADAM12}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020164-T01-IM6-28	P-0020164-T01-IM6	P-0020164	Hepatobiliary Cancer	IHCH	FGFR2-ADAM9	ADAM9	in-frame	five_prime	123242738	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ADAM9 (NM_003816) fusion: t(8;10(p11.22;q26.13)(chr8:g.38901161::chr10:g.123242738) | Protein Fusion: in frame {FGFR2:ADAM9}	NA
EVT-P-0018920-T01-IM6-29	P-0018920-T01-IM6	P-0018920	Cancer of Unknown Primary	ADNOS	FGFR2-AHCYL1	AHCYL1	in-frame	five_prime	123242706	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - AHCYL1 (NM_006621) fusion (FGFR2 exons 1-17 fused in-frame with AHCYL1 exons 2-17): t(1;10)(p13.3;q26.13)(chr1:g.110531185::chr10:g.123242706) | Protein Fusion: in frame {FGFR2:AHCYL1}	NA
EVT-P-0000644-T01-IM3-30	P-0000644-T01-IM3	P-0000644	Hepatobiliary Cancer	IHCH	FGFR2-AHCYL1	AHCYL1	in-frame	three_prime	123239745	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		NA | Protein fusion: in frame (AHCYL1-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012666-T01-IM5-31	P-0012666-T01-IM5	P-0012666	Hepatobiliary Cancer	IHCH	FGFR2-AMOT	AMOT	in-frame	five_prime	123241762	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - AMOT (NM_001113490) Rearrangement: t(10;X)(q26.13.1;q23)(chr10:g.123241762::chrX:g.112058354) | Protein fusion: in frame (FGFR2-AMOT)	NA
EVT-P-0047504-T01-IM6-32	P-0047504-T01-IM6	P-0047504	Breast Cancer	IDC	FGFR2-ATAD1	ATAD1	in-frame	five_prime	123241680	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ATAD1 (NM_032810) fusion: c.2301+1532:FGFR2_c.831+3993:ATAD1del | Protein Fusion: in frame {FGFR2:ATAD1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017424-T01-IM6-33	P-0017424-T01-IM6	P-0017424	Endometrial Cancer	USC	FGFR2-ATE1	ATE1	out-of-frame	five_prime	123240443	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.2302-908:FGFR2_c.1379-520:ATE1dup | Protein Fusion: out of frame {FGFR2:ATE1}	NA
EVT-P-0063512-T01-IM7-36	P-0063512-T01-IM7	P-0063512	Prostate Cancer	PRAD	FGFR2-ATE1	ATE1	unknown	five_prime	123263336	10	469	False	469	True	exact match	lost	0.0	False		Protein kinase domain; C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.1407:FGFR2_c.943-1893:ATE1dup | Protein Fusion: mid-exon {FGFR2:ATE1}	NA
EVT-P-0063625-T01-IM7-37	P-0063625-T01-IM7	P-0063625	Endometrial Cancer	USC	FGFR2-ATE1	ATE1	in-frame	five_prime	123242847	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_c.975+6106:ATE1dup | Protein Fusion: in frame {FGFR2:ATE1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059296-T01-IM7-38	P-0059296-T01-IM7	P-0059296	Cancer of Unknown Primary	ADNOS	FGFR2-BAIAP2L1	BAIAP2L1	in-frame	five_prime	123242027	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98029133::chr10:g.123242027) | Protein Fusion: in frame {FGFR2:BAIAP2L1}	NA
EVT-P-0052794-T01-IM6-39	P-0052794-T01-IM6	P-0052794	Hepatobiliary Cancer	CHOL	FGFR2-BAIAP2L1	BAIAP2L1	in-frame	five_prime	123240813	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::chr10:g.123240813) | Protein Fusion: in frame {FGFR2:BAIAP2L1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006885-T01-IM5-41	P-0006885-T01-IM5	P-0006885	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240587	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) rearrangement: c.2302-1052:FGFR2_c.2388-5239:BICC1inv | Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0049133-T04-IM6-42	P-0049133-T04-IM6	P-0049133	Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123242098	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+1114:FGFR2_c.237+29879:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021327-T01-IM6-43	P-0021327-T01-IM6	P-0021327	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240344	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21): c.2302-1613:FGFR2_c.2377+32254:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0013174-T01-IM5-44	P-0013174-T01-IM5	P-0013174	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240123	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2302-588:FGFR2_c.2182-284:BICC1inv | Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039087-T01-IM6-45	P-0039087-T01-IM6	P-0039087	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123242059	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2301+1153:FGFR2_c.2181+133:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0049133-T02-IM6-46	P-0049133-T02-IM6	P-0049133	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123242098	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+1114:FGFR2_c.237+29879:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049135-T03-IM6-47	P-0049135-T03-IM6	P-0049135	Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123239791	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-256:FGFR2_c.238-36290:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0049135-T07-IM7-48	P-0049135-T07-IM7	P-0049135	Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123239791	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-256:FGFR2_c.238-36290:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0051773-T01-IM6-49	P-0051773-T01-IM6	P-0051773	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123242828	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+384:FGFR2_c.237+13961:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053822-T01-IM6-50	P-0053822-T01-IM6	P-0053822	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123243033	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+179:FGFR2_c.238-1932:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0060245-T01-IM7-51	P-0060245-T01-IM7	P-0060245	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	unknown	five_prime	123239377	18	820	False	820	True	exact match	retained	1.0	False	Protein kinase domain		C-terminal autoinhibitory tail (exon 18)	FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2460:FGFR2_c.1179+1103:BICC1inv | Protein Fusion: mid-exon {FGFR2:BICC1}	NA
EVT-P-0068033-T01-IM7-53	P-0068033-T01-IM7	P-0068033	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123239673	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-138:FGFR2_c.238-40422:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068640-T01-IM7-54	P-0068640-T01-IM7	P-0068640	Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123240444	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-909:FGFR2_c.238-18504:BICC1inv | Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0004520-T01-IM5-55	P-0004520-T01-IM5	P-0004520	Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240269	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2(NM_000141) - BICC1(NM_001080512) Fusion (FGFR2 exon 17 fused with exon 10 of BICC1) : c.2302-734:FGFR2_c.1180-924:BICC1inv | Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054191-T01-IM6-56	P-0054191-T01-IM6	P-0054191	Hepatobiliary Cancer	IHCH	FGFR2-BICD1	BICD1	in-frame	five_prime	123242849	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BICD1 (NM_001714) fusion: t(10;12)(q26.13;p11.21)(chr10:g.123242849::chr12:g.32465127) | Protein Fusion: in frame {FGFR2:BICD1}	NA
EVT-P-0061764-T01-IM7-57	P-0061764-T01-IM7	P-0061764	Cancer of Unknown Primary	ADNOS	FGFR2-BNIP2	BNIP2	in-frame	five_prime	123243170	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BNIP2 (NM_004330) fusion: t(10;15)(q26.13;q22.2)(chr10:g.123243170::chr15:g.59975666) | Protein Fusion: in frame {FGFR2:BNIP2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020135-T01-IM6-58	P-0020135-T01-IM6	P-0020135	Colorectal Cancer	COAD	FGFR2-BTBD16	BTBD16	out-of-frame	five_prime	123239702	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - BTBD16 (NM_144587) rearrangement: c.2302-167:FGFR2_c.386-107: BTBD16inv | Protein Fusion: out of frame {FGFR2:BTBD16}	NA
EVT-P-0033652-T01-IM6-59	P-0033652-T01-IM6	P-0033652	Non-Small Cell Lung Cancer	LUAD	FGFR2-CCDC186	CCDC186	in-frame	five_prime	123240830	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - CCDC186 (NM_018017) Rearrangement : c.2302-1295:FGFR2_c.1222-3356:CCDC186del | Protein Fusion: in frame {FGFR2:CCDC186}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060582-T01-IM7-60	P-0060582-T01-IM7	P-0060582	Cancer of Unknown Primary	ADNOS	FGFR2-CCDC73	CCDC73	in-frame	five_prime	123241248	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - CCDC73 (NM_001008391) fusion: t(10;11)(q26.13;p13)(chr10:g.123241248::chr11:g.32747346) | Protein Fusion: in frame {FGFR2:CCDC73}	NA
EVT-P-0037574-T01-IM6-61	P-0037574-T01-IM6	P-0037574	Glioma	GNG	FGFR2-CTNNA3	CTNNA3	in-frame	five_prime	123240911	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - CTNNA3 (NM_013266) fusion: c.2302-1376:FGFR2_c.1884+30653:CTNNA3del | Protein Fusion: in frame {FGFR2:CTNNA3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044768-T01-IM6-63	P-0044768-T01-IM6	P-0044768	Breast Cancer	IDC	FGFR2-DIABLO	DIABLO	in-frame	five_prime	123243152	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - DIABLO (NM_019887) fusion: t(10;12)(q26.13;q24.31)(chr10:g.123243152::chr12:g.122708063) | Protein Fusion: in frame {FGFR2:DIABLO}	NA
EVT-P-0061783-T01-IM7-100	P-0061783-T01-IM7	P-0061783	Pancreatic Cancer	PAAD	FGFR2-FOXP1	FOXP1	in-frame	five_prime	123241483	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(chr3:g.71192573::chr10:g.123241483) | Protein Fusion: in frame {FGFR2:FOXP1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058539-T01-IM6-101	P-0058539-T01-IM6	P-0058539	Colorectal Cancer	COAD	FGFR2-FOXP1	FOXP1	unknown	five_prime	123239434	18	801	False	801	True	exact match	retained	1.0	False	Protein kinase domain		C-terminal autoinhibitory tail (exon 18)	FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(chr3:g.71113500::chr10:g.123239434) | Protein Fusion: mid-exon {FGFR2:FOXP1}	NA
EVT-P-0055215-T01-IM6-103	P-0055215-T01-IM6	P-0055215	Ovarian Cancer	HGSOC	FGFR2-GRID1	GRID1	out-of-frame	five_prime	123323934	4	152	True	151	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain; C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - GRID1 (NM_017551) rearrangement: c.454+82:FGFR2_c.726+103661:GRID1del | Protein Fusion: out of frame {FGFR2:GRID1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015437-T01-IM6-104	P-0015437-T01-IM6	P-0015437	Breast Cancer	IDC	FGFR2-HSPA12A	HSPA12A	in-frame	five_prime	123242591	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - HSPA12A (NM_025015) rearrangement: c.2301+621:FGFR2_c.663+3346:HSPA12Adel | Protein fusion: in frame (FGFR2-HSPA12A)	NA
EVT-P-0009701-T01-IM5-105	P-0009701-T01-IM5	P-0009701	Cancer of Unknown Primary	CUP	FGFR2-KCNH7	KCNH7	unknown	five_prime	123239444	18	798	False	797	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain		C-terminal autoinhibitory tail (exon 18)	FGFR2 (NM_000141) - KCNH7 (NM_033272) Rearrangement : t(2,10)(q24.3;q26.13) (chr2:g.163263969::chr10:g.123239444) | Protein fusion: mid-exon (FGFR2-KCNH7)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013405-T01-IM5-107	P-0013405-T01-IM5	P-0013405	Hepatobiliary Cancer	IHCH	FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123241766	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c.2301+1446:FGFR2_c.354+73179:KIAA1217inv | Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-0008658-T01-IM5-108	P-0008658-T01-IM5	P-0008658	Hepatobiliary Cancer	IHCH	FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123241193	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c.2302-1658:FGFR2_c.355-22274:KIAA1217inv | Protein fusion: in frame (FGFR2-KIAA1217)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034784-T01-IM6-109	P-0034784-T01-IM6	P-0034784	Hepatobiliary Cancer	IHCH	FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123242254	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - KIAA1217 (NM_019590) fusion: c.2301+958:FGFR2_c.355-49659:KIAA1217inv | Protein Fusion: in frame {FGFR2:KIAA1217}	NA
EVT-P-0003660-T01-IM5-110	P-0003660-T01-IM5	P-0003660	Cancer of Unknown Primary	CUP	FGFR2-KIF14	KIF14	in-frame	five_prime	123240038	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		t(1;10)(q32.1;q26.13)(chr1:g.200555931::chr10:g.123240038) | Protein fusion: in frame (FGFR2-KIF14)	NA
EVT-P-0013241-T01-IM5-111	P-0013241-T01-IM5	P-0013241	Hepatobiliary Cancer	IHCH	FGFR2-KLHL29	KLHL29	unknown	three_prime	123241923	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		KLHL29 (NM_052920) - FGFR2 (NM_000141) rearrangement: t(2;10)(p24.1;q26.13)(chr2:g.:23647979::chr10:g.123241923) | Transcript fusion (KLHL29-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019439-T02-IM6-112	P-0019439-T02-IM6	P-0019439	Hepatobiliary Cancer	HCCIHCH	FGFR2-L3MBTL3	L3MBTL3	in-frame	five_prime	123239680	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - L3MBTL3 (NM_032438) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::chr10:g.123239680) | Protein Fusion: in frame {FGFR2:L3MBTL3}	NA
EVT-P-0011394-T01-IM5-113	P-0011394-T01-IM5	P-0011394	Hepatobiliary Cancer	IHCH	FGFR2-LAMC1	LAMC1	in-frame	five_prime	123240042	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - LAMC1 (NM_002293) rearrangment: t(1;10)(q25.3;q26.13)(chr1:g.183109471::chr10:g.123240042) | Protein fusion: in frame (FGFR2-LAMC1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064855-T01-IM7-115	P-0064855-T01-IM7	P-0064855	Non-Small Cell Lung Cancer	LUAD	FGFR2-MBIP	MBIP	in-frame	five_prime	123240070	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - MBIP (NM_016586) fusion: t(10;14)(q26.13;q13.3)(chr10:g.123240070::chr14:g.36787130) | Protein Fusion: in frame {FGFR2:MBIP}	NA
EVT-P-0030844-T01-IM6-116	P-0030844-T01-IM6	P-0030844	Colorectal Cancer	COAD	FGFR2-MYH15	MYH15	in-frame	five_prime	123240665	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - MYH15 (NM_014981) rearrangement: t(3;10)(q13.13;q26.13)(chr3:g.108134575::chr10:g.123240665) | Protein Fusion: in frame {FGFR2:MYH15}	NA
EVT-P-0027623-T01-IM6-117	P-0027623-T01-IM6	P-0027623	Hepatobiliary Cancer	IHCH	FGFR2-MYLK	MYLK	unknown	five_prime	123240959	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) rearrangement: t(3;10)(q21.1;q26.13)(chr3:g.123525612::chr10:g.123240959) | Transcript Fusion { FGFR2:MYLK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012423-T02-IM6-118	P-0012423-T02-IM6	P-0012423	Hepatobiliary Cancer	EHCH	FGFR2-NOL4	NOL4	in-frame	five_prime	123242055	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.123242055::chr18:g.31563418) | Protein Fusion: in frame {FGFR2::NOL4}	NA
EVT-P-0038235-T01-IM6-119	P-0038235-T01-IM6	P-0038235	Hepatobiliary Cancer	IHCH	FGFR2-NOL4	NOL4	in-frame	five_prime	123240418	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t(10;18)(q26.13;q12.1)(chr10:g.123240418::chr18:g.31580817) | Protein Fusion: in frame {FGFR2::NOL4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006328-T02-IM6-120	P-0006328-T02-IM6	P-0006328	Hepatobiliary Cancer	IHCH	FGFR2-NRBF2	NRBF2	in-frame	five_prime	123241246	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157-363NRBF2inv | Protein Fusion: in frame {FGFR2:NRBF2}	NA
EVT-P-0058339-T01-IM6-121	P-0058339-T01-IM6	P-0058339	Cancer of Unknown Primary	CUPNOS	FGFR2-PAH	PAH	in-frame	five_prime	123240026	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026::chr12:g.103294473) | Protein Fusion: in frame {FGFR2:PAH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015500-T01-IM6-122	P-0015500-T01-IM6	P-0015500	Cancer of Unknown Primary	CUP	FGFR2-PALLD	PALLD	in-frame	five_prime	123239869	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - PALLD (NM_001166108) Rearrangement: t(4,10)(q33,q26.13)(chr4:g.169605121::chr10:g.123239869) | Protein fusion: in frame (FGFR2-PALLD)	NA
EVT-P-0021448-T03-IM6-123	P-0021448-T03-IM6	P-0021448	Non-Small Cell Lung Cancer	SARCL	FGFR2-PKP4	PKP4	in-frame	five_prime	123242661	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - PKP4 (NM_003628) rearrangement: t(2;10)(q24.1;q26.13)(chr2:g.159395413::chr10:g.123242661) | Protein Fusion: in frame {FGFR2::PKP4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059669-T01-IM7-125	P-0059669-T01-IM7	P-0059669	Hepatobiliary Cancer	IHCH	FGFR2-POC1B	POC1B	in-frame	five_prime	123242798	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - POC1B (NM_001199777.2) fusion: t(10;12)(q26.13;q21.33)(chr10:g.123242798::chr12:g.89857934) | Protein Fusion: in frame {FGFR2:POC1B}	NA
EVT-P-0008657-T01-IM5-126	P-0008657-T01-IM5	P-0008657	Non-Small Cell Lung Cancer	LUAD	FGFR2-PPP1R21	PPP1R21	in-frame	five_prime	123240684	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR1 (NM_000141) - PPP1R21 (NM_152994) Rearrangement : t(2;10)(2p16.3;10q26.13)(chr2:g.48721864::chr10:g.123240684) | Protein fusion: in frame (FGFR2-KLRAQ1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057247-T02-IM7-127	P-0057247-T02-IM7	P-0057247	Glioma	GB	FGFR2-PRDX3	PRDX3	in-frame	five_prime	123242894	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - PRDX3 (NM_006793) fusion: c.2301+318:FGFR2_c.36+287:PRDX3del | Protein Fusion: in frame {FGFR2:PRDX3}	NA
EVT-P-0011394-T01-IM5-129	P-0011394-T01-IM5	P-0011394	Hepatobiliary Cancer	IHCH	FGFR2-RABGAP1L	RABGAP1L	in-frame	five_prime	123240065	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - RABGAP1L (NM_014857) rearrangement: t(1;10)(q25.1;q26.13)(chr1:g.174863117::chr10:g.123240065) | Protein fusion: in frame (FGFR2-RABGAP1L)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012639-T01-IM5-130	P-0012639-T01-IM5	P-0012639	Hepatobiliary Cancer	IHCH	FGFR2-SEPTIN10	SEPTIN10	unknown	five_prime	123240798	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - SEPT10 (NM_144710) rearrangement: t(2;10)(q13;q26.13)(chr2:g.110325395::chr10:g.123240798) | Protein fusion: mid-exon (FGFR2-SEPT10)	NA
EVT-P-0055202-T01-IM6-131	P-0055202-T01-IM6	P-0055202	Hepatobiliary Cancer	IHCH	FGFR2-SH3GLB1	SH3GLB1	in-frame	five_prime	123242019	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - SH3GLB1 (NM_001206651) rearrangement: t(1;10)(p22.3;q26.13)(chr1:g.87191292::chr10:g.123242019) | Protein Fusion: in frame {FGFR2:SH3GLB1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011516-T01-IM5-132	P-0011516-T01-IM5	P-0011516	Hepatobiliary Cancer	IHCH	FGFR2-SHC2	SHC2	in-frame	five_prime	123243182	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - SH2C (NM_012435) Rearrangement : t(10;19)(q26.13;p13.3)(chr10:g.123243182::chr19:g.447779) | Protein fusion: in frame (FGFR2-SHC2)	NA
EVT-P-0057005-T01-IM6-133	P-0057005-T01-IM6	P-0057005	Esophagogastric Cancer	EGC	FGFR2-SHTN1	SHTN1	in-frame	five_prime	123239901	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2302-366:FGFR2_c.534+660:KIAA1598del | Protein Fusion: in frame {FGFR2:KIAA1598}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032152-T01-IM6-134	P-0032152-T01-IM6	P-0032152	Pancreatic Cancer	PAAD	FGFR2-SHTN1	SHTN1	in-frame	five_prime	123241513	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2301+1699:FGFR2_c.534+751:KIAA1598 | Protein Fusion: in frame {FGFR2:KIAA1598}	NA
EVT-P-0064573-T01-IM7-135	P-0064573-T01-IM7	P-0064573	Endometrial Cancer	UEC	FGFR2-SKI	SKI	in-frame	five_prime	123242232	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - SKI (NM_003036) fusion: t(1;10)(p36.33;q26.13)(chr1:g.2207811::chr10:g.123242232) | Protein Fusion: in frame {FGFR2:SKI}	NA
EVT-P-0054244-T01-IM6-136	P-0054244-T01-IM6	P-0054244	Cancer of Unknown Primary	CUP	FGFR2-SORBS1	SORBS1	in-frame	five_prime	123243201	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - SORBS1 (NM_001034954) fusion: c.2301+11:FGFR2_c.1738-382:SORBS1del | Protein Fusion: in frame {FGFR2:SORBS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057277-T01-IM6-137	P-0057277-T01-IM6	P-0057277	Hepatobiliary Cancer	IHCH	FGFR2-SRRT	SRRT	out-of-frame	five_prime	123242651	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - SRRT (NM_015908) fusion: t(7;10)(q22.1;q26.13)(chr7:g.100475766::chr10:g.123242651) | Protein Fusion: out of frame {FGFR2:SRRT}	NA
EVT-P-0055975-T01-IM6-139	P-0055975-T01-IM6	P-0055975	Esophagogastric Cancer	ESCA	FGFR2-STAT4	STAT4	in-frame	five_prime	123240312	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - STAT4 (NM_001243835) rearrangement: t(2;10)(q32.3;q26.13)(chr2:g.191936121::chr10:g.123240312) | Protein Fusion: in frame {FGFR2:STAT4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057746-T01-IM6-140	P-0057746-T01-IM6	P-0057746	Hepatobiliary Cancer	CHOL	FGFR2-TACC2	TACC2	in-frame	five_prime	123241998	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128-402:TACC2inv | Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-0057746-T02-IM6-141	P-0057746-T02-IM6	P-0057746	Hepatobiliary Cancer	IHCH	FGFR2-TACC2	TACC2	in-frame	five_prime	123241998	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128-402:TACC2inv | Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-0045185-T01-IM6-142	P-0045185-T01-IM6	P-0045185	Colorectal Cancer	COAD	FGFR2-TACC2	TACC2	in-frame	five_prime	123240014	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2302-479:FGFR2_c.8128-2249:TACC2inv | Protein Fusion: in frame {FGFR2:TACC2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036358-T01-IM6-143	P-0036358-T01-IM6	P-0036358	Hepatobiliary Cancer	IHCH	FGFR2-TACC2	TACC2	in-frame	five_prime	123239743	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2302-208:FGFR2_c.7344+468:TACC2inv | Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-0024187-T01-IM6-145	P-0024187-T01-IM6	P-0024187	Hepatobiliary Cancer	IHCH	FGFR2-TACC2	TACC2	unknown	five_prime	123240573	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TACC2 (NM_206862) Fusion (FGFR2 exon 17 with TACC2 exon12) : c.2302-1038:FGFR2_c.8009:TACC2inv | Protein Fusion: mid-exon {FGFR2:TACC2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031242-T01-IM6-146	P-0031242-T01-IM6	P-0031242	Breast Cancer	BRCA	FGFR2-TACC2	TACC2	unknown	five_prime	123239432	18	802	False	801	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain		C-terminal autoinhibitory tail (exon 18)	FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2405:FGFR2_c.7896+237:TACC2inv | Protein Fusion: mid-exon {FGFR2:TACC2}	NA
EVT-P-0050962-T01-IM6-148	P-0050962-T01-IM6	P-0050962	Cervical Cancer	CEAS	FGFR2-TACC2	TACC2	in-frame	five_prime	123241844	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1368:FGFR2_c.7608+332:TACC2inv | Protein Fusion: in frame {FGFR2:TACC2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023364-T03-IM6-150	P-0023364-T03-IM6	P-0023364	Colorectal Cancer	COAD	FGFR2-TACC2	TACC2	unknown	five_prime	123241587	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TACC2 (NM_206862) rearrangement: c.2301+1625:FGFR2_c.34-5469:TACC2dup | Antisense Fusion	NA
EVT-P-0022337-T06-IM6-151	P-0022337-T06-IM6	P-0022337	Breast Cancer	IDC	FGFR2-TBC1D4	TBC1D4	in-frame	five_prime	123241723	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946350) | Protein Fusion: in frame {FGFR2:TBC1D4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022337-T04-IM6-152	P-0022337-T04-IM6	P-0022337	Breast Cancer	BRCA	FGFR2-TBC1D4	TBC1D4	in-frame	five_prime	123241723	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946350) | Protein Fusion: in frame {FGFR2:TBC1D4}	NA
EVT-P-0005291-T01-IM5-153	P-0005291-T01-IM5	P-0005291	Hepatobiliary Cancer	IHCH	FGFR2-TFEC	TFEC	unknown	five_prime	123241729	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_022970) - TFEC (NM_012252) Rearrangement: t(10;7)(q26.13;q31.2)(chr10:g.123241729;chr7:g.115644112) | Transcript fusion (FGFR2-TFEC)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051558-T01-IM6-154	P-0051558-T01-IM6	P-0051558	Hepatobiliary Cancer	CHOL	FGFR2-TTC28	TTC28	in-frame	five_prime	123243158	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TTC28 (NM_001145418) fusion: t(10;22)(q26.13;q12.1)(chr10:g.123243158::chr22:g.28605071) | Protein Fusion: in frame {FGFR2:TTC28}	NA
EVT-P-0016640-T01-IM6-155	P-0016640-T01-IM6	P-0016640	Hepatobiliary Cancer	CHOL	FGFR2-TTC28	TTC28	in-frame	five_prime	123241944	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000142) - TTC28 (NM_001145418) Rearrangement: t(10;22)(q26.11;q12.1)(chr10:g.123241944::chr22:g.28666001) | Protein Fusion: in frame {FGFR2:TTC28}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029726-T01-IM6-156	P-0029726-T01-IM6	P-0029726	Breast Cancer	IDC	FGFR2-UBP1	UBP1	in-frame	five_prime	123240062	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - UBP1 (NM_001128161) fusion: t(3;10)(p22.3;q26.13)(chr3:g.33449450::chr10:g.123240062) | Protein Fusion: in frame {FGFR2:UBP1}	NA
EVT-P-0004089-T01-IM5-158	P-0004089-T01-IM5	P-0004089	Cancer of Unknown Primary	CUPNOS	FGFR2-VCL	VCL	unknown	five_prime	123242749	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2(NM_000141) - VCL (NM_014000) Rearrangement : c.2301+463:FGFR2_c.1032:VCLinv | Protein fusion: mid-exon (FGFR2-VCL)	NA
EVT-P-0059238-T02-IM7-159	P-0059238-T02-IM7	P-0059238	Hepatobiliary Cancer	CHOL	FGFR2-WAC	WAC	in-frame	five_prime	123242066	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301+1146:FGFR2_c.1437+679:WACinv | Protein Fusion: in frame {FGFR2:WAC}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058031-T01-IM6-160	P-0058031-T01-IM6	P-0058031	Hepatobiliary Cancer	IHCH	FGFR2-WAC	WAC	in-frame	five_prime	123243180	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301+32:FGFR2_c.382-351:WACinv | Protein Fusion: in frame {FGFR2:WAC}	NA
EVT-P-0025807-T01-IM6-161	P-0025807-T01-IM6	P-0025807	Hepatobiliary Cancer	IHCH	FGFR2-YTHDF3	YTHDF3	unknown	three_prime	123242317	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		YTHDF3 (NM_152758) - FGFR2 (NM_000141) rearrangement: t(8;10)(q12.3;q26.13)(chr8:g.64081967::chr10:g.123242317) | Protein Fusion: mid-exon {YTHDF3:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0056399-T01-IM6-162	P-0056399-T01-IM6	P-0056399	Esophagogastric Cancer	ESCA	FGFR2-ZFAT	ZFAT	in-frame	five_prime	123240282	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ZFAT (NM_020863) rearrangement: t(8;10)(q24.22;q26.13)(chr8:g.135564146::chr10:g.123240282) | Protein Fusion: in frame {FGFR2:ZFAT}	NA
EVT-P-0042241-T01-IM6-163	P-0042241-T01-IM6	P-0042241	Hepatobiliary Cancer	IHCH	FGFR2-ZMYM4	ZMYM4	in-frame	five_prime	123242889	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ZMYM4 (NM_005095) fusion: t(1;10)(p34.3;q26.13)(chr1:g.35833702::chr10:g.123242889) | Protein Fusion: in frame {FGFR2:ZMYM4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016195-T01-IM6-164	P-0016195-T01-IM6	P-0016195	Hepatobiliary Cancer	IHCH	GAB2-FGFR2	GAB2	in-frame	five_prime	123242812	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - GAB2 (NM_080491) fusion (FGFR2 exons 1-17 fused with GAB2 exons 2-10) : t(10;11)(q26.11;q14.1)(chr10:g.123242812::chr11:g.78030553) | Protein Fusion: in frame {FGFR2:GAB2}	NA
EVT-P-0013756-T03-IM6-165	P-0013756-T03-IM6	P-0013756	Cancer of Unknown Primary	CUP	INA-FGFR2	INA	in-frame	three_prime	123242695	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		INA (NM_032727) - FGFR2 (NM_000141) rearrangement: c.1066-3500:INA_c.2301+517:FGFR2inv | Protein Fusion: in frame {INA:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013756-T01-IM5-166	P-0013756-T01-IM5	P-0013756	Cancer of Unknown Primary	CUP	INA-FGFR2	INA	in-frame	three_prime	123242695	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		INA (NM_032727) - FGFR2 (NM_000141) Rearrangement : c.1066-3500:INA_c.2301+517:FGFR2inv | Protein fusion: in frame (INA-FGFR2)	NA
EVT-P-0002810-T01-IM3-169	P-0002810-T01-IM3	P-0002810	Ampullary Cancer	PAMPCA	KIAA1217-FGFR2	KIAA1217	in-frame	five_prime	123240874	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		NA | Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-0000386-T02-IM5-170	P-0000386-T02-IM5	P-0000386	Bladder Cancer	BLCA	MARVELD3-FGFR2	MARVELD3	unknown	five_prime	123243101	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		MARVELD3 (NM_001017967) - FGFR2 (NM_000141) rearrangement: t(16;10)(16q22.2;10q26.13) (chr16:g.71660650::chr10:123243101) | Antisense fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024094-T01-IM6-172	P-0024094-T01-IM6	P-0024094	Hepatobiliary Cancer	IHCH	MYH9-FGFR2	MYH9	in-frame	five_prime	123240418	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - MYH9 (NM_002473) rearrangement: t(10;22)(q26.13;q12.3)(chr10:g.123240418::chr22:g.36682296) | Protein Fusion: in frame {FGFR2:MYH9}	NA
EVT-P-0032208-T01-IM6-173	P-0032208-T01-IM6	P-0032208	Hepatobiliary Cancer	IHCH	MYO18B-FGFR2	MYO18B	in-frame	five_prime	123241929	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - MYO18B (NM_032608) rearrangement: t(10;22)(q26.13;q12.2)(chr10:g.123241929::chr22:g.26280218) | Protein Fusion: in frame {FGFR2:MYO18B}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021975-T01-IM6-175	P-0021975-T01-IM6	P-0021975	Breast Cancer	IDC	NID2-FGFR2	NID2	out-of-frame	three_prime	123240637	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		NID2 (NM_007361) - FGFR2 (NM_000141) rearrangement: t(10;14)(q26.13;q22.1)(chr10:g.123240637::chr14:g.52482743) | Protein Fusion: out of frame {NID2:FGFR2}	NA
EVT-P-0012423-T01-IM5-176	P-0012423-T01-IM5	P-0012423	Hepatobiliary Cancer	CHOL	NOL4-FGFR2	NOL4	in-frame	five_prime	123242054	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - NOL4 (NM_003787) Rearrangement :t(10;18)(q26.13;q12.1)(chr10:g.123242054::chr18:g.31563418) | Protein fusion: in frame (FGFR2-NOL4)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023103-T01-IM6-177	P-0023103-T01-IM6	P-0023103	Hepatobiliary Cancer	IHCH	NOL4-FGFR2	NOL4	in-frame	five_prime	123241921	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused to NOL4 exons 10-11): t(10;18)(q26.13;q12.1)(chr10:g.123241921::chr18:g.31518701) | Protein Fusion: in frame {FGFR2::NOL4}	NA
EVT-P-0008693-T01-IM5-178	P-0008693-T01-IM5	P-0008693	Hepatobiliary Cancer	IHCH	NRAP-FGFR2	NRAP	in-frame	five_prime	123242277	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - NRAP (NM_198060) rearrangement: c.2301+935:FGFR2_c.2458-427:NRAPdel | Protein fusion: in frame (FGFR2-NRAP)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006328-T01-IM5-179	P-0006328-T01-IM5	P-0006328	Hepatobiliary Cancer	IHCH	NRBF2-FGFR2	NRBF2	in-frame	five_prime	123241245	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - NRBF2 (NM_030759) rearrangement: c.2302-1710:FGFR2_c.157-364:NRBF2inv | Protein fusion: in frame (FGFR2-NRBF2)	NA
EVT-P-0049071-T01-IM6-180	P-0049071-T01-IM6	P-0049071	Non-Small Cell Lung Cancer	LUSC	NRG3-FGFR2	NRG3	out-of-frame	three_prime	123244882	17	732	True	732	True	exact match	disrupted	0.093862815884847654	True	C-terminal autoinhibitory tail (exon 18)		Protein kinase domain	NRG3 (NM_001010848) - FGFR2 (NM_000141) rearrangement: c.823+108972:NRG3_c.2195+27:FGFR2inv | Protein Fusion: out of frame {NRG3:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022337-T01-IM6-181	P-0022337-T01-IM6	P-0022337	Breast Cancer	IDC	NRG3-FGFR2	NRG3	out-of-frame	three_prime	123241723	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		NRG3 (NM_001010848) - FGFR2 (NM_000141) Rearrangement : c.953+89877:NRG3_c.2301+1489:FGFR2inv | Protein Fusion: out of frame {NRG3:FGFR2}	NA
EVT-P-0032171-T03-IM6-182	P-0032171-T03-IM6	P-0032171	Ovarian Cancer	LGSOC	OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv | Protein Fusion: in frame {FGFR2:OPTN}	NA
EVT-P-0032171-T01-IM6-183	P-0032171-T01-IM6	P-0032171	Ovarian Cancer	LGSOC	OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv | Protein Fusion: in frame {FGFR2:OPTN}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022781-T01-IM6-184	P-0022781-T01-IM6	P-0022781	Ovarian Cancer	HGSOC	PDZRN4-FGFR2	PDZRN4	in-frame	five_prime	123241997	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::chr10:g.123239680) | Protein Fusion: in frame {FGFR2:PDZRN4}	NA
EVT-P-0044768-T01-IM6-187	P-0044768-T01-IM6	P-0044768	Breast Cancer	IDC	RNF145-FGFR2	RNF145	out-of-frame	three_prime	123243127	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		RNF145 (NM_144726) - FGFR2 (NM_000141) rearrangement: t(5;10)(q33.3;q26.13)(chr5:g.158598146::chr10:g.123243127) | Protein Fusion: out of frame {RNF145:FGFR2}	NA

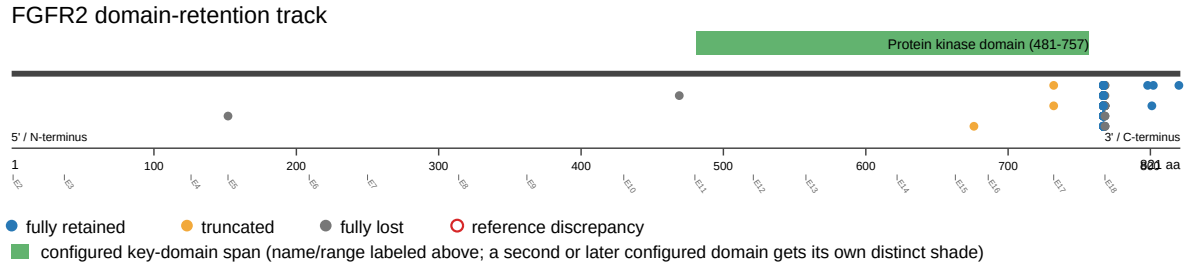
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006235-T01-IM5-188	P-0006235-T01-IM5	P-0006235	Hepatobiliary Cancer	IHCH	ROCK1-FGFR2	ROCK1	in-frame	five_prime	123240133	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - ROCK1 (NM_005406) rearrangement: t(10;18)(q26.13;q11.1)(chr10:g.123240133::chr18:g18680051) | Protein fusion: in frame (FGFR2-ROCK1)	NA
EVT-P-0039370-T01-IM6-189	P-0039370-T01-IM6	P-0039370	Hepatobiliary Cancer	IHCH	RPAP3-FGFR2	RPAP3	in-frame	three_prime	123242832	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		RPAP3 (NM_024604) - FGFR2 (NM_000141) rearrangement: t(10;12)(q26.13;q13.11)(chr10:g.123242832::chr12:g.48092574) | Protein Fusion: in frame {RPAP3:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025074-T01-IM6-190	P-0025074-T01-IM6	P-0025074	Cancer of Unknown Primary	ADNOS	SHTN1-FGFR2	SHTN1	in-frame	five_prime	123240122	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - KIAA1598 (NM_001127211) Fusion (FGFR2 exon 17 with KIAA1598 exon 5): c.2302-587:FGFR2_c.267+2596:KIAA1598 | Protein Fusion: in frame {FGFR2:KIAA1598}	NA
EVT-P-0032507-T01-IM6-191	P-0032507-T01-IM6	P-0032507	Hepatobiliary Cancer	IHCH	SORBS1-FGFR2	SORBS1	unknown	five_prime	123242268	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) Rearrangement : c.2301+944:FGFR2_chr10:g.97250518del | Transcript Fusion { FGFR2:SORBS1 }	NA

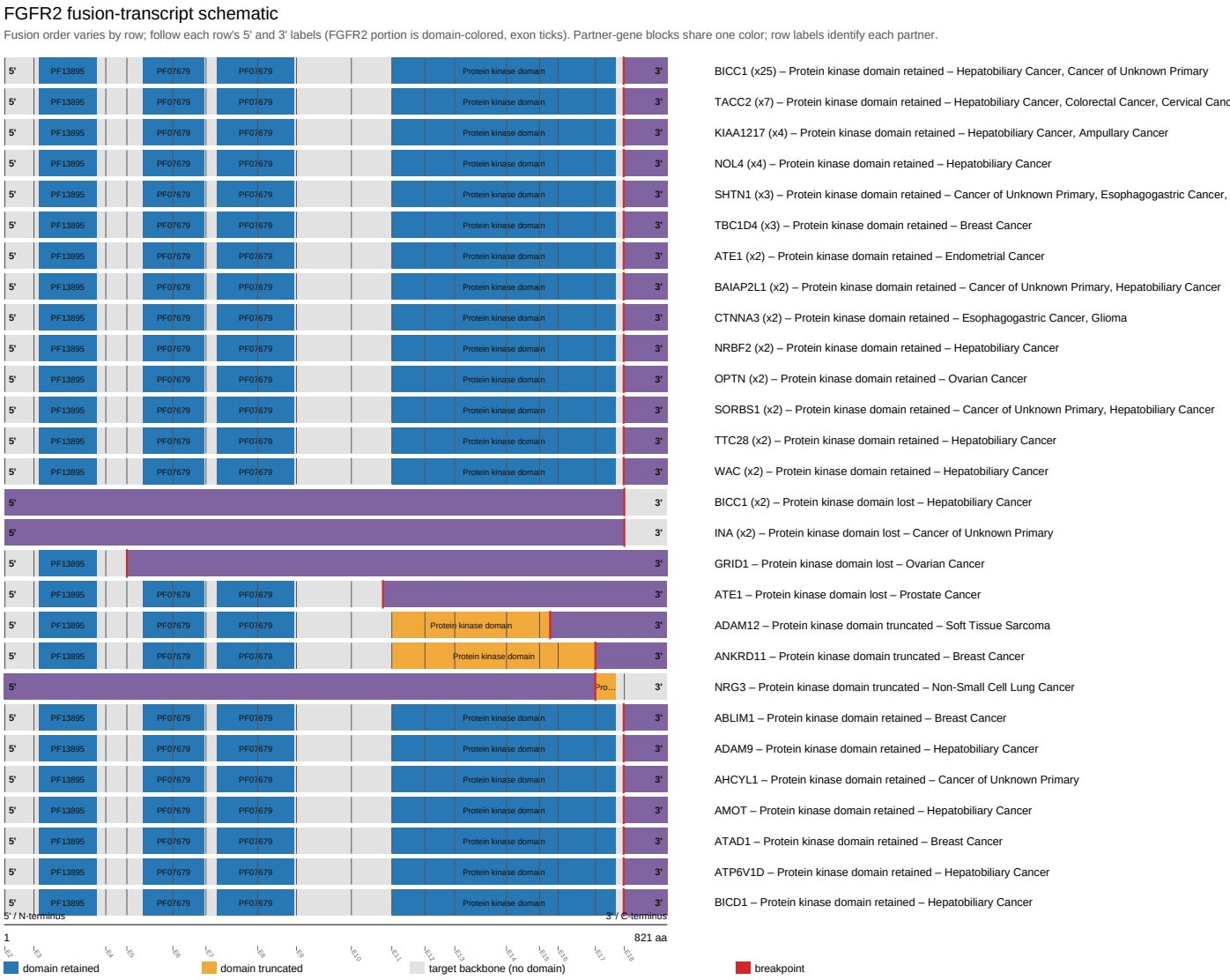
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022337-T01-IM6-192	P-0022337-T01-IM6	P-0022337	Breast Cancer	IDC	TBC1D4-FGFR2	TBC1D4	in-frame	five_prime	123241723	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		FGFR2 (NM_000141) - TBC1D4 (NM_014832) Rearrangement : t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946348) | Protein Fusion: in frame {FGFR2:TBC1D4}	NA
EVT-P-0021788-T01-IM6-193	P-0021788-T01-IM6	P-0021788	Breast Cancer	IDC	VPS26A-FGFR2	VPS26A	in-frame	three_prime	123242226	18	768	True	768	True	exact match	lost	0.0	False	C-terminal autoinhibitory tail (exon 18)	Protein kinase domain		VPS26A (NM_004896) - FGFR2 (NM_000141) Rearrangement : c.4-1035:VPS26A_c.2301+986:FGFR2inv | Protein Fusion: in frame {VPS26A:FGFR2}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 84 partner/breakpoint groups by recurrence; see results.tsv for the complete list.